St. Philomena's College (Autonomous),
Bannimantap, Mysuru - 15.

PG Department of Mathematics

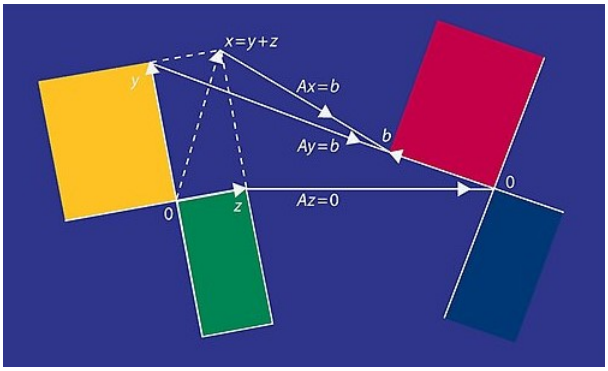
A report on extension activity conducted in connection with cracking higher mathematics competitive exams

held on 23-08-2022

Crash Course on Linear Algebra - *Basic ideas and MCQ tackling Techniques*

Resource Person: Mr. Bharatha K,
Assistant Professor,
PG Department of Mathematics,
St. Philomena's College (Autonomous), Mysuru.

Venue: Room No. 79, PG Block.



Objectives

1) To motivate the students to clear higher mathematics exams.

2) To express the strategies and techniques involved in solving MCQ's.

3) To facilitate the students by creating an exposure towards problem solving skills.

4) To strengthen students mathematical ideas by illustrative examples

Outlines of the Session:

• Overview of Linear Algebra.

• Basic ideas of Vector Spaces, properties, Subspaces.

• Results and consequences of linear spaces, linear combinations and spans.

• Ideas from Linear transformations, Eigenvalues, Eigenvectors, Inner product spaces, Bilinear transformations.

• Interaction.



PG Department of Mathematics of St. Philomena's College, Mysuru conducted this crash course with a purpose to help students to clear higher mathematics exams, in order to persue potential teaching or research in recognized institutes. Mr. Puneeth S, Assistant Professor of the department welcomed the august gathering and requested PG Director to give a message to the students. Prof. Orthbert Pinto, PG Director, stressed the importance and the opportunities in pursuing research by clearing competitive exams.

The resource person Mr. Bharatha K. started his session by recollecting few important and significant aspects of Linear Algebra in connection with Physics, Algorithms and Mathematical Computations.

He listed the preliminaries of Linear Algebra and the definitions. He explained Vector spaces, Basis elements, properties of vector spaces, Subspaces, Linear dependence, independence and Linear Span. Further, he illustrated few basic examples and made the students enough confident in the subject of context.On continuation, he posed few tricky questions and explained the advanced results that are needed to solve such questions in non-cumbersome way.





He briefly explained the important results of Eigenvalues, Eigenvectors, rank, nullity and linear transformations. Further, he discussed the techniques such as option elimination, counterexamples, verification, use of suitable definitions and results were taught. In the ensuing hour, he solved the MCQ questions appeared in CSIR NET, SLET, GATE, NBHM, TIFR etc. and explained the tricks and strategies involved in answering MCQ's.

Moving on, he gave basic ideas from Inner product spaces, Bilinear Transformations, Orthogonal Projections, Orthogonal Spaces.

He solved pertinent questions that stressed the deeper understanding of the subject in every aspects. He took questions with multiple correct options and explained the tackling techniques to solve them. Assignments was given to the students and it led to one to one interactions. After that interactive session, he explained the time management and gave suggestions in choosing pure as well as applied concepts according to weightage of marks. He listed the reference books and the online platforms that serves the purpose.



Finally, he thanked the management of St. Philomena's College and the students for their active participation.